SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Fluid Mechanics & Hydraulic Machinery: Bernoulli & Turbines

Department: MECHANICAL	**Course / Subject:** Fluid Mechanics & Hydraulic Machinery (ME401PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Fluid statics (Manometry), Continuity, Momentum, Bernoulli's energy equation, Laminar & Turbulent pipe flow, Pelton wheel, Francis and Kaplan turbines.

Unit 1: Fluid Statics & Kinematics

Fluid properties (Viscosity, Surface tension, Capillarity), Pascal's law, Hydrostatic force on submerged planes, Streamlines, Streaklines, Continuity equation.

Unit 2: Fluid Dynamics

Euler's equation of motion along a streamline, Bernoulli's energy theorem derivation, Venturimeter, Orifice meter, Pitot tube measurements, Momentum equation.

Unit 3: Flow Through Pipes

Reynolds experiment, Laminar flow between parallel plates (Couette flow), Hagen-Poiseuille pipe flow, Darcy-Weisbach friction factor equation, Moody chart.

Unit 4: Hydraulic Impulse Turbines

Impact of fluid jets on flat and curved vanes, Pelton wheel component construction, Velocity triangles, Hydraulic efficiency, Mechanical and overall efficiency.

Unit 5: Reaction Turbines & Centrifugal Pumps

Francis turbine, Kaplan axial flow turbine, Draft tube theory, Cavitation phenomenon, Centrifugal pump working, Manometric head, Specific speed of pumps.